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# Analog MEMS Electronic Stethoscope

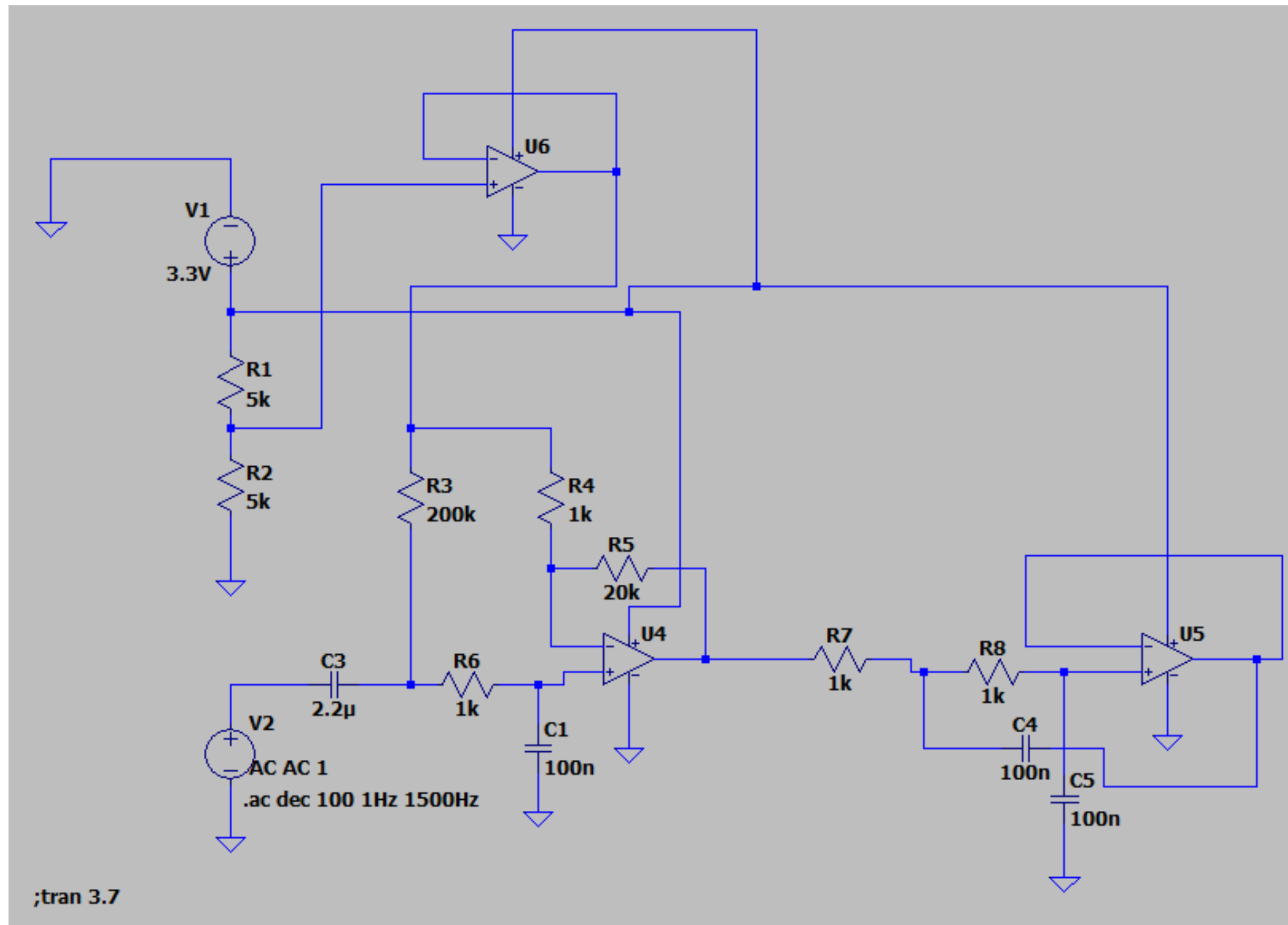
Simulation , PCB & Enclosure Designs

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# LTspice Simulation of the PCG Signal Conditioning Circuit



## What This Simulation Shows

- **Test signal:** V2 supplies an AC test excitation (1 Hz–1.5 kHz sweep) representing the raw PCG (heart sound) waveform at the circuit input.
- **Purpose:** LTspice simulation is used to verify the conditioning circuit's gain, filtering and biasing behaviour before any hardware is built.
- **Signal path:** Input → AC coupling and re-biasing onto a stable 1.65 V reference → inverting amplifier stage (U4) → two-stage RC low-pass filter → buffered output stage (U5).
- **What is verified:** Correct signal amplification, attenuation of unwanted high-frequency content, and a stable single-supply reference voltage generated by R1/R2 and buffered by U6.
- **Outcome:** The output waveform preserves the shape of the PCG envelope at a much larger, ADC-compatible amplitude, confirming the circuit design before physical construction.

PCG Input  
(V2)



Coupling + 1.65V  
Bias (C3, R1/R2,  
U6)



Amplification  
(U4, approx.  
x20)



Low-Pass  
Filtering  
(R7/R8, C4/C5)

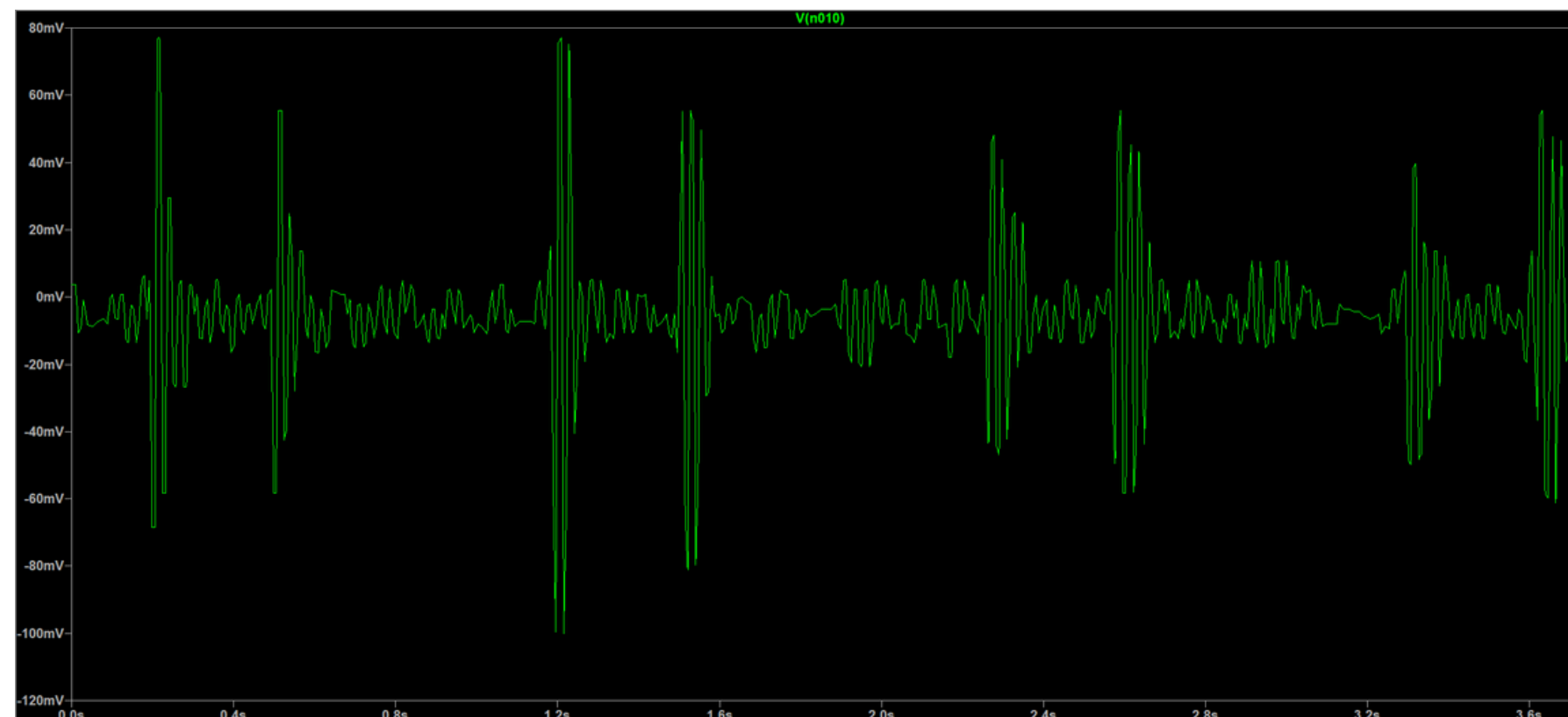


Buffered  
Output  
(U5)



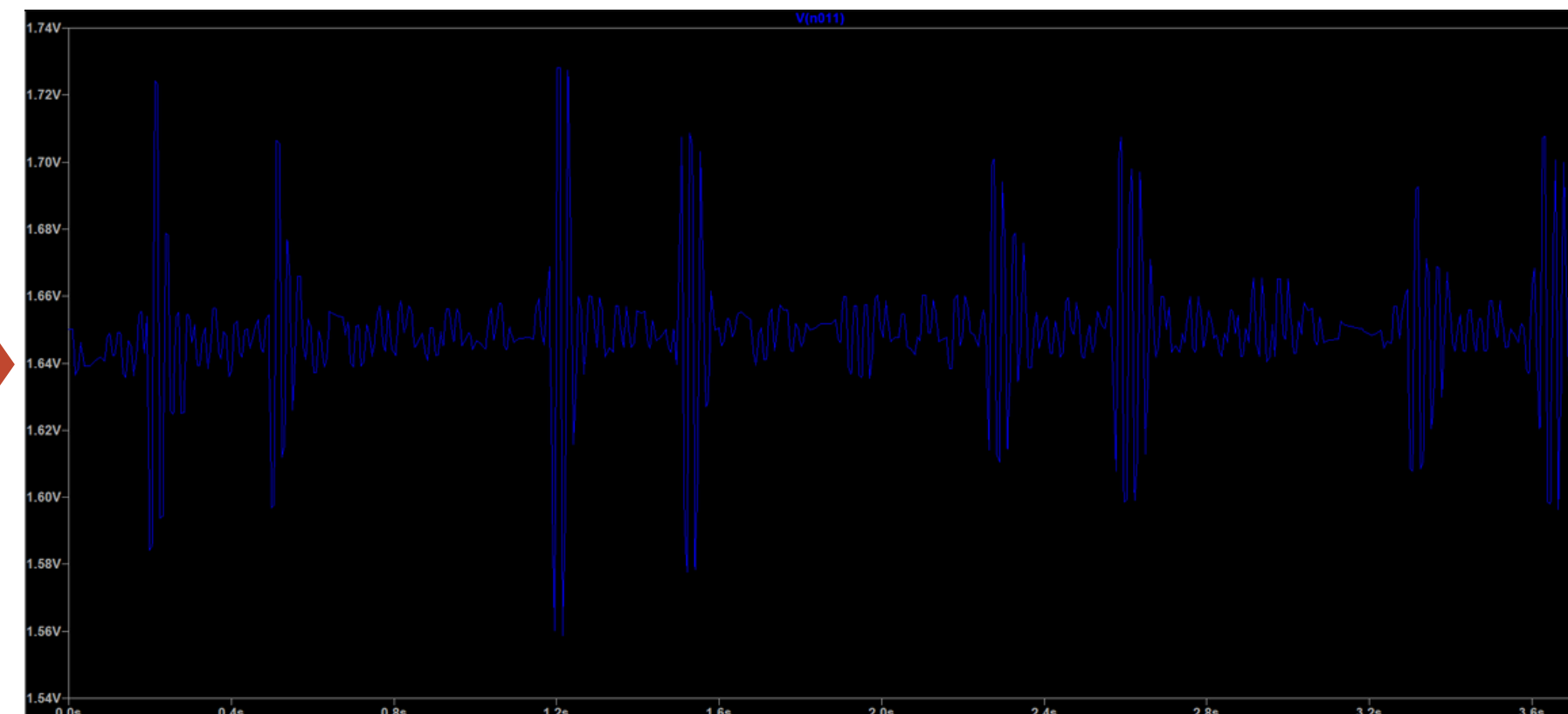
# Real-World Signal and Noise Modelling

## Original PCG Input Signal (node n010)



Raw AC test signal from source V2 — small amplitude, centred at 0 V, with realistic high-frequency fluctuation already present in the recorded waveform.

## Signal After Coupling + Bias (node n011)

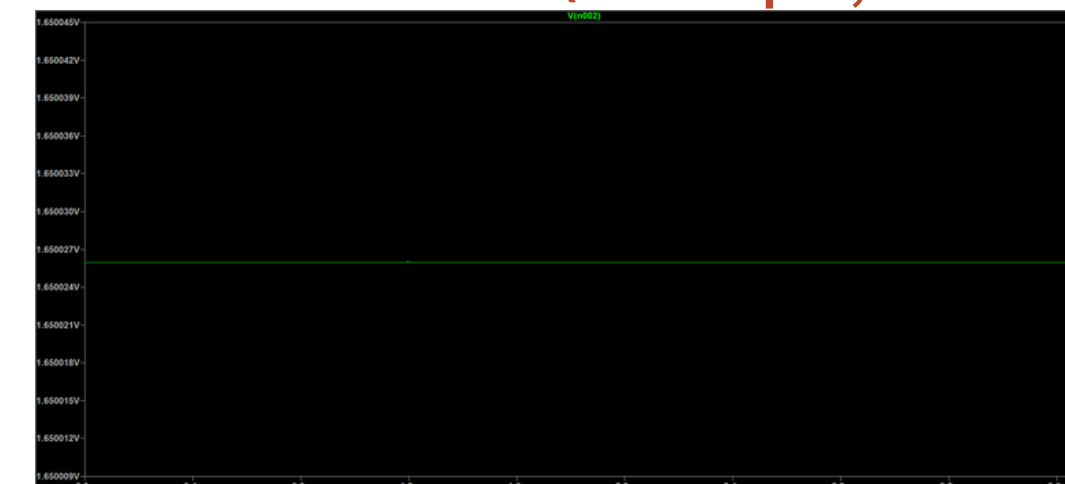


Same waveform after capacitor C3 removes the DC component and re-centres the signal onto the 1.65 V reference, ready for single-supply amplification.

## Why This Matters

- **Realistic excitation:** The fluctuation visible in the input waveform is part of the recorded/simulated PCG signal itself; no separate power-line or motion-artifact source is present in this circuit.
- **Bias stage:** Coupling capacitor C3 blocks the DC level of V2 and re-references the AC signal onto the 1.65 V rail, generated by divider R1/R2 and buffered by U6 for a low-impedance, stable reference.
- **Why simulate this first:** Confirming that a realistic, imperfect input signal is correctly biased and passed through before amplification reduces the risk of clipping, distortion, or reference drift once the circuit is built in hardware.

## Buffered Reference (U6 Output, node

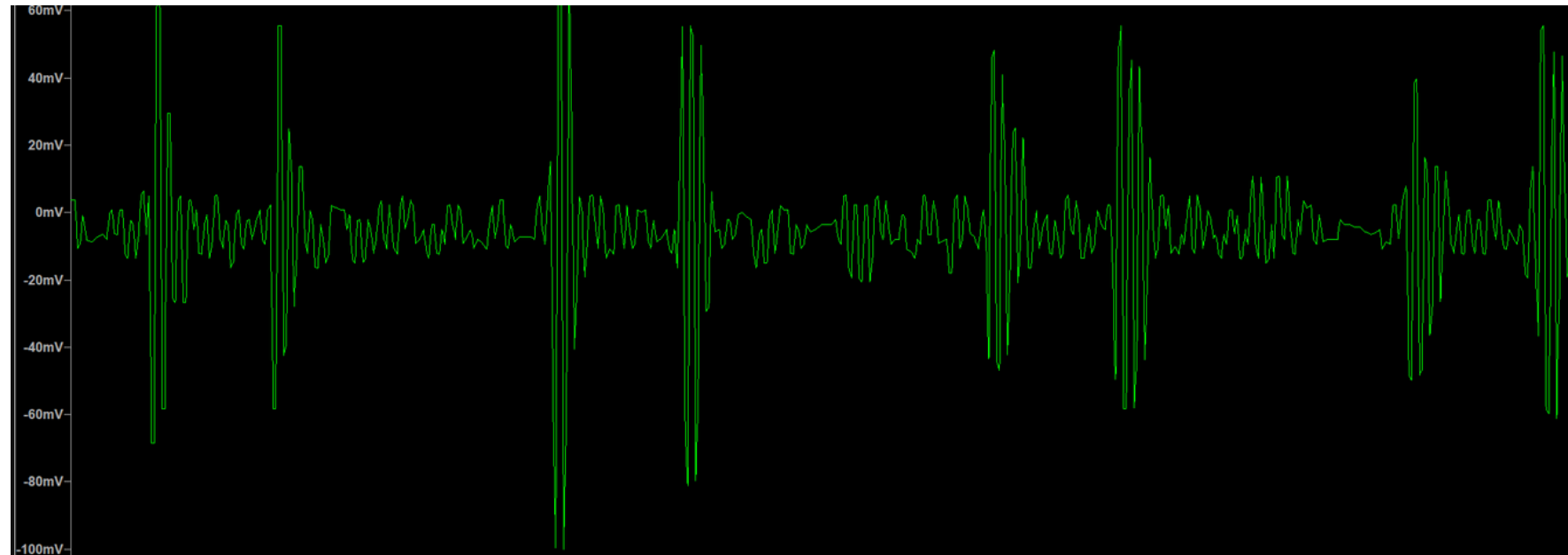


Steady at approx. 1.65 V — the stable bias the input signal is coupled onto.



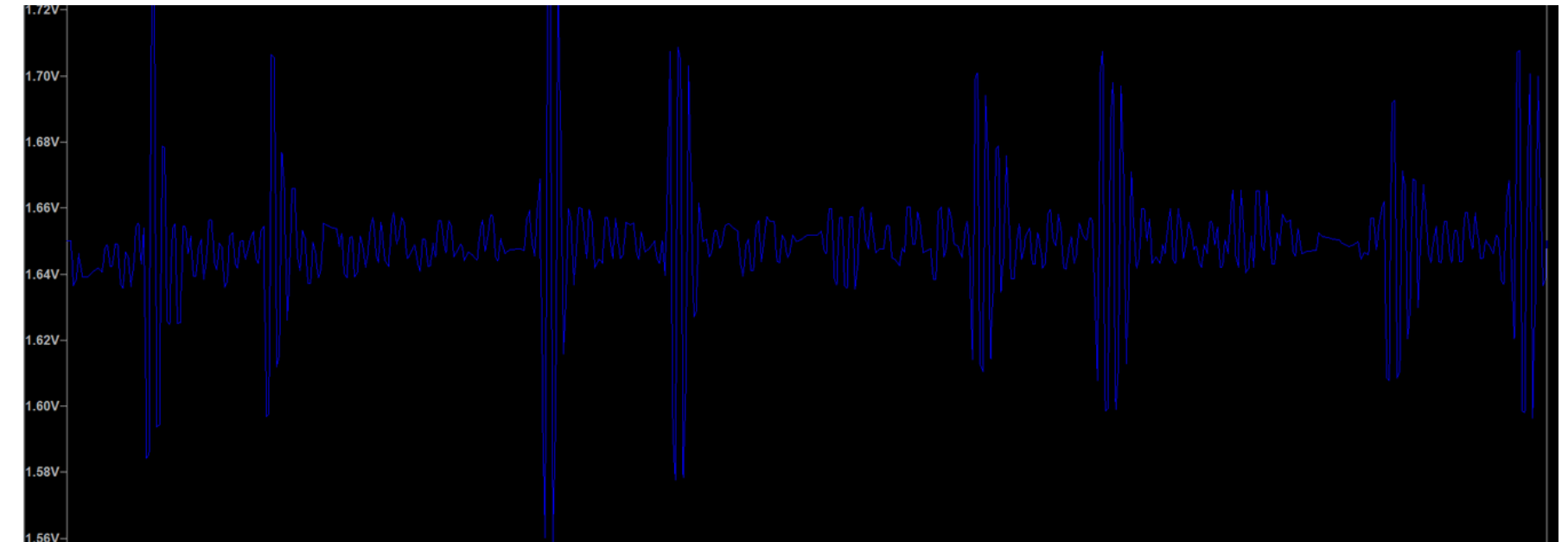
# Signal Transformation Through the Circuit

## 1. Input Signal (n010)



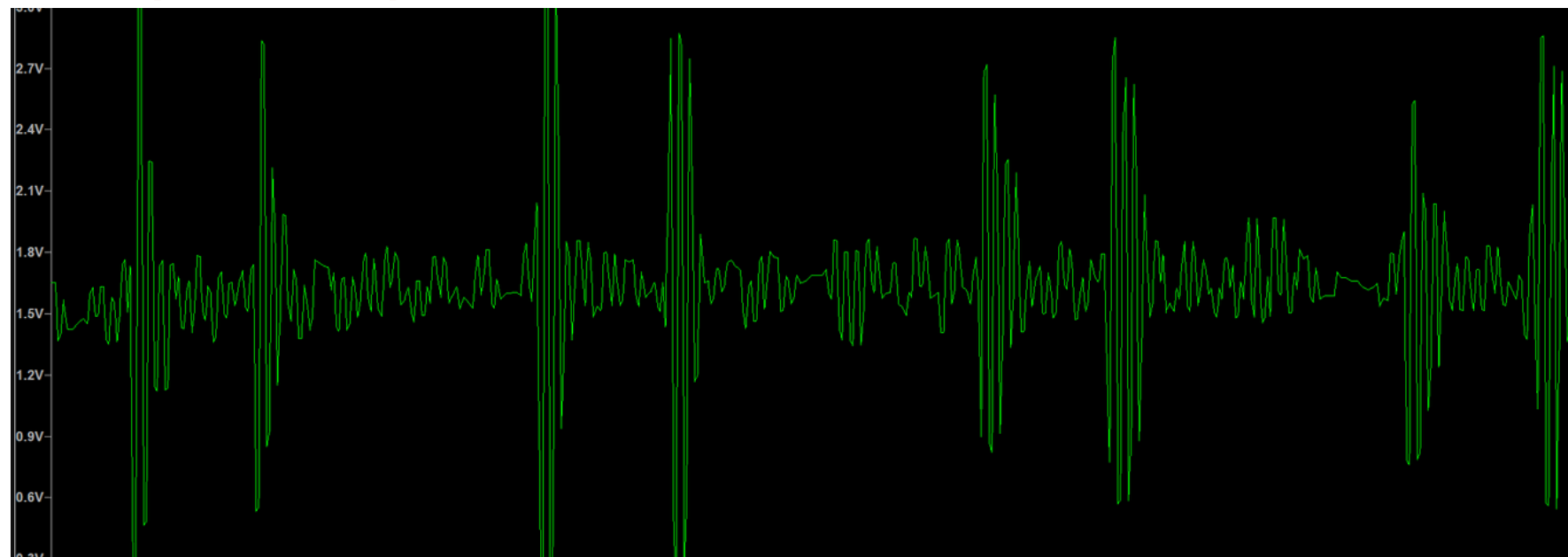
Raw PCG-like test signal entering the circuit from V2, small amplitude, centred at 0 V.

## 2. After Coupling + Bias (n011)



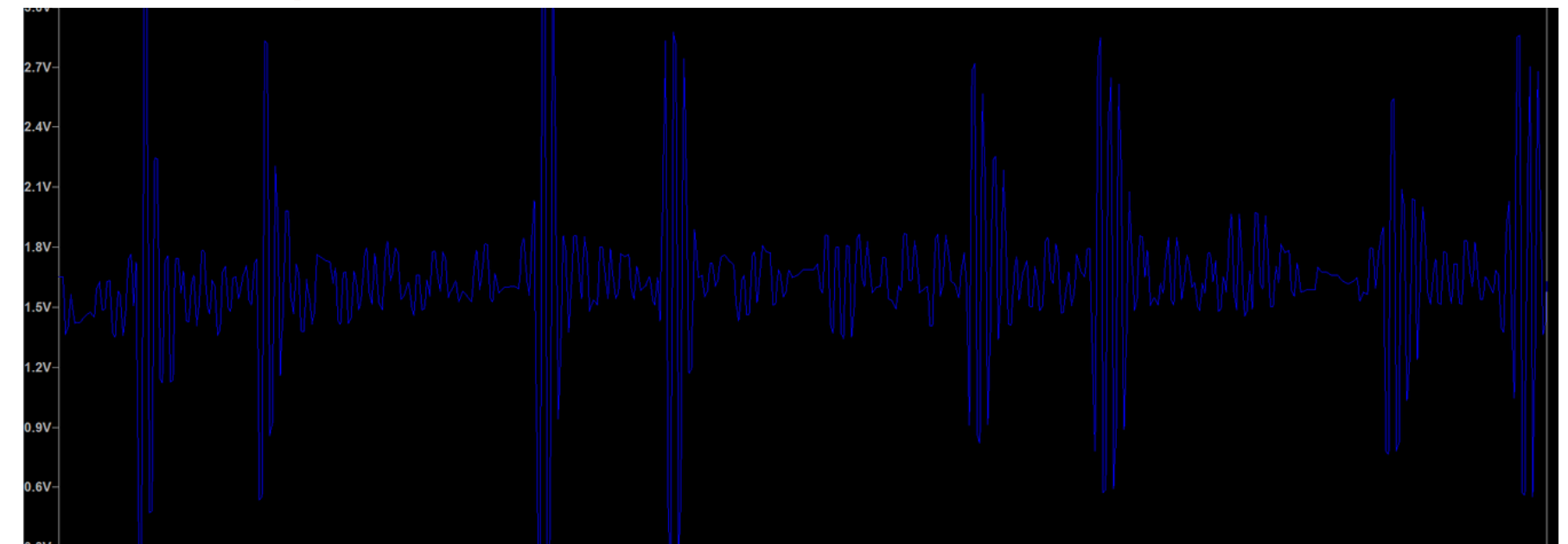
Signal after C3 removes the DC level and re-centres it onto the 1.65 V reference at the U4 input.

## 3. Amplifier Output (n006)



Amplitude increases substantially — the U4 inverting stage applies a gain of roughly x20 ( $R5/R4$ ).

## 4. Final Output (n004)



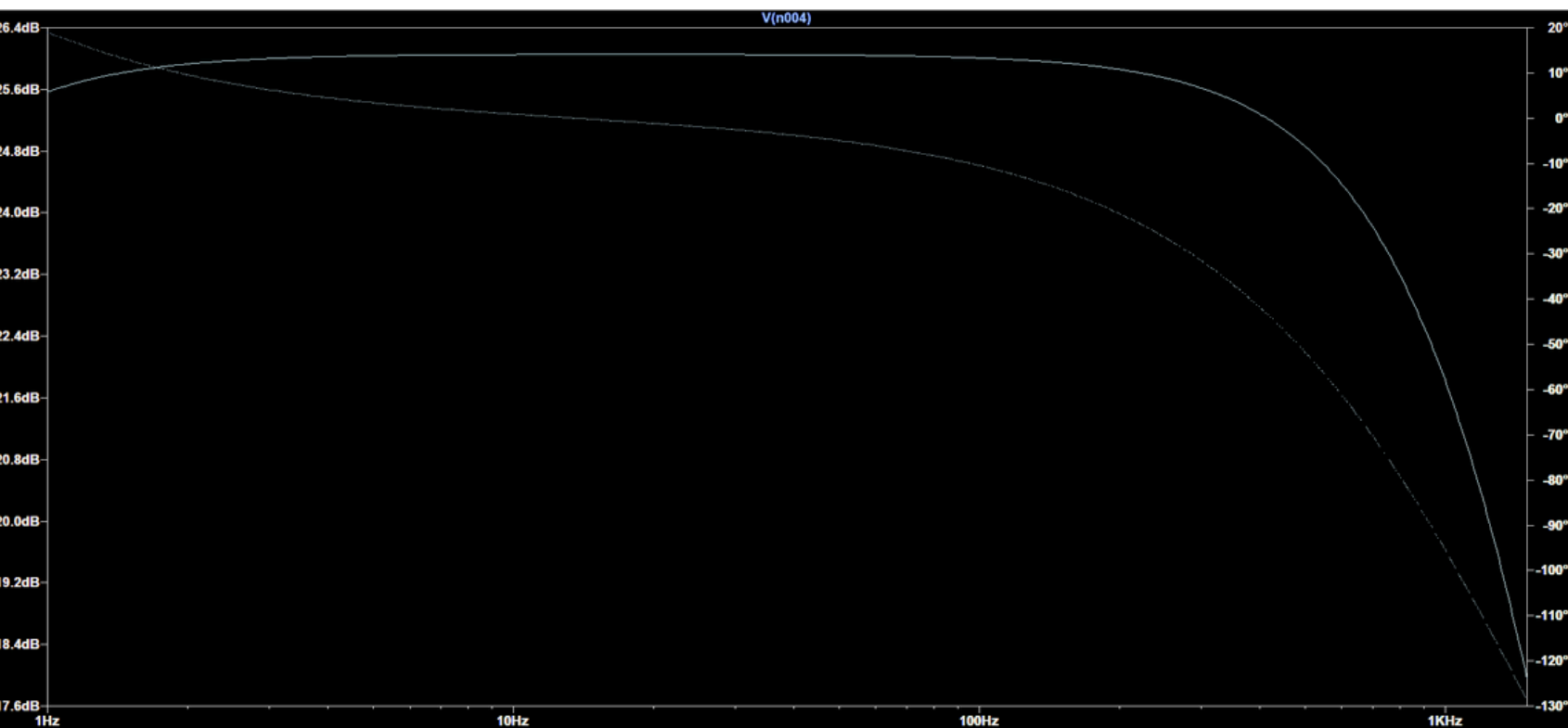
After RC low-pass filtering and buffering by U5: a larger, cleaner, ADC-ready conditioned PCG signal.



# Simulation Results and Signal Quality Improvement

| Parameter            | Input / Raw Signal                                 | Final Output                                                          |
|----------------------|----------------------------------------------------|-----------------------------------------------------------------------|
| Signal amplitude     | approx. $\pm 100\text{mV}$ , centred at 0 V (n010) | approx. 0.6 V – 3.3 V, centred at 1.65 V (n004)                       |
| Reference / baseline | No fixed bias (raw AC source)                      | Stable 1.65 V reference (n002, n003)                                  |
| Frequency response   | Full-band input (1 Hz – 1.5 kHz sweep)             | approx. 26 dB flat gain, 1 Hz to approx. 600 Hz; rolls off above that |
| Signal quality       | Low-level, unconditioned                           | Amplified (approx. $\times 20$ ) and band-limited, ADC-ready          |

## Frequency Response – Output V(n004)



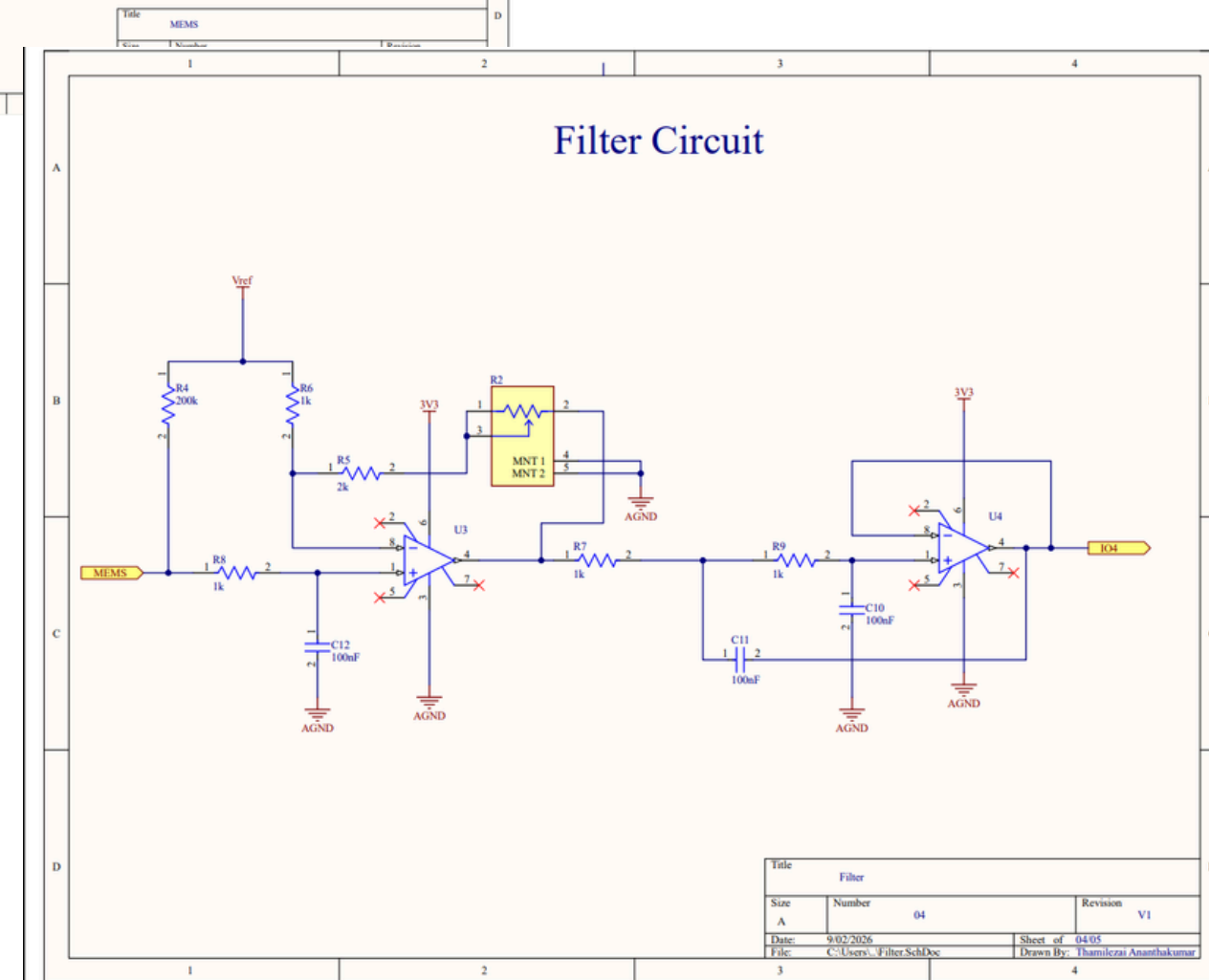
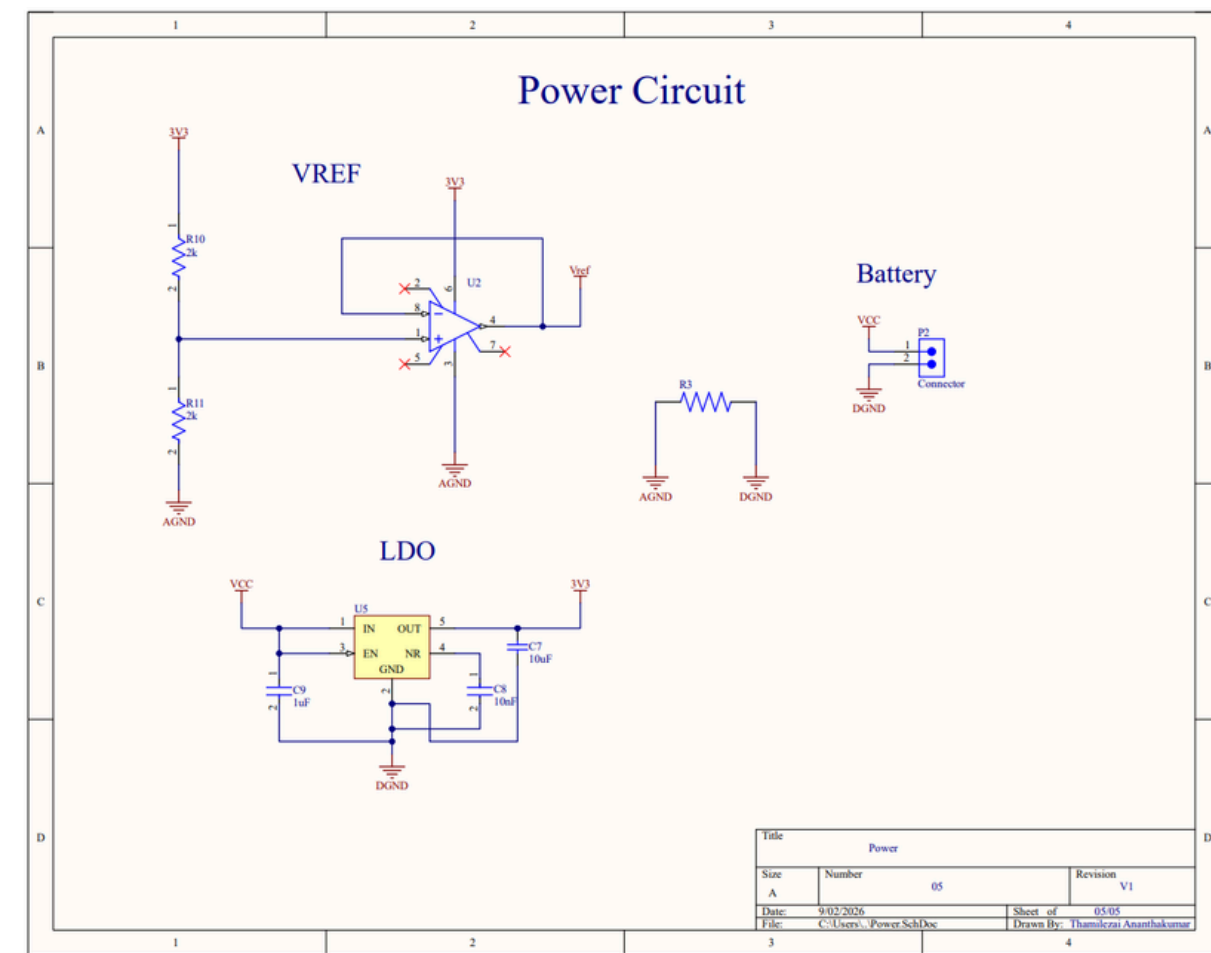
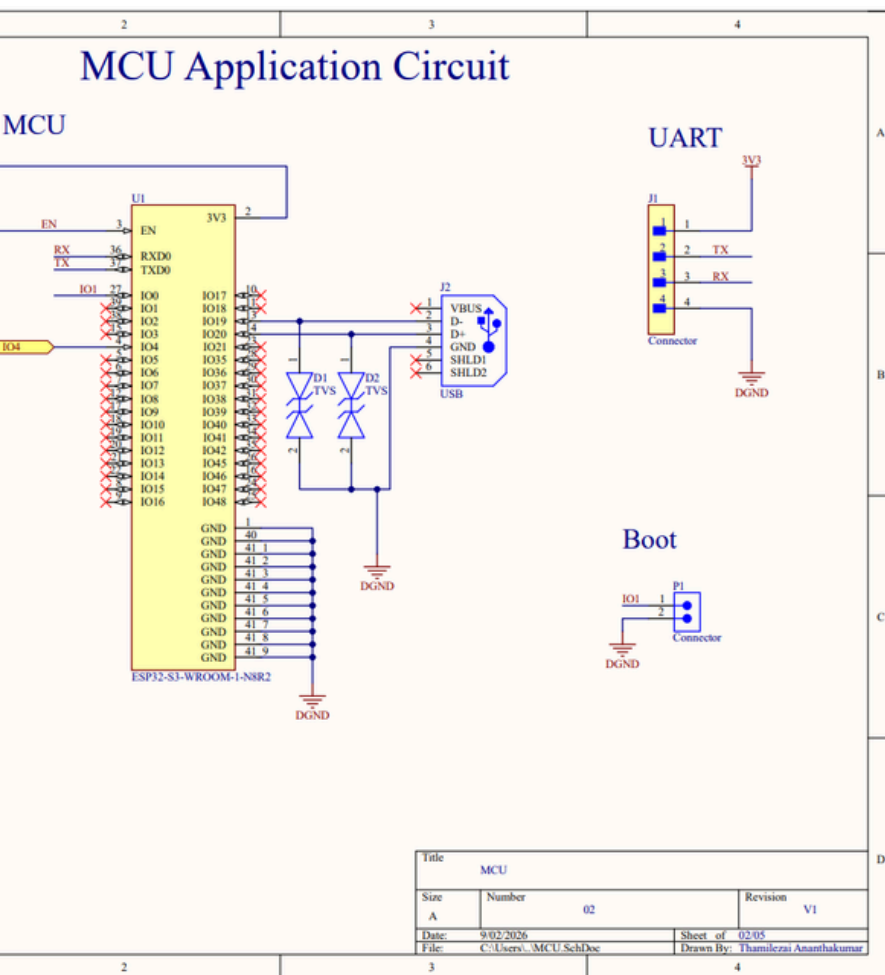
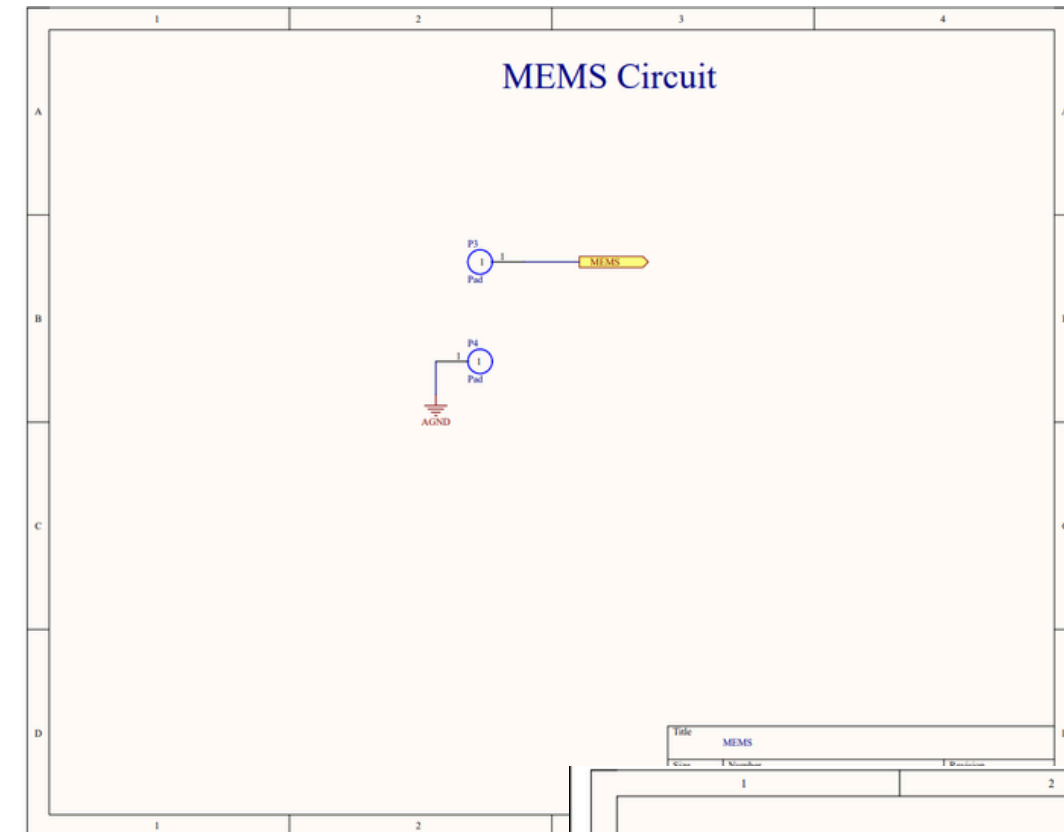
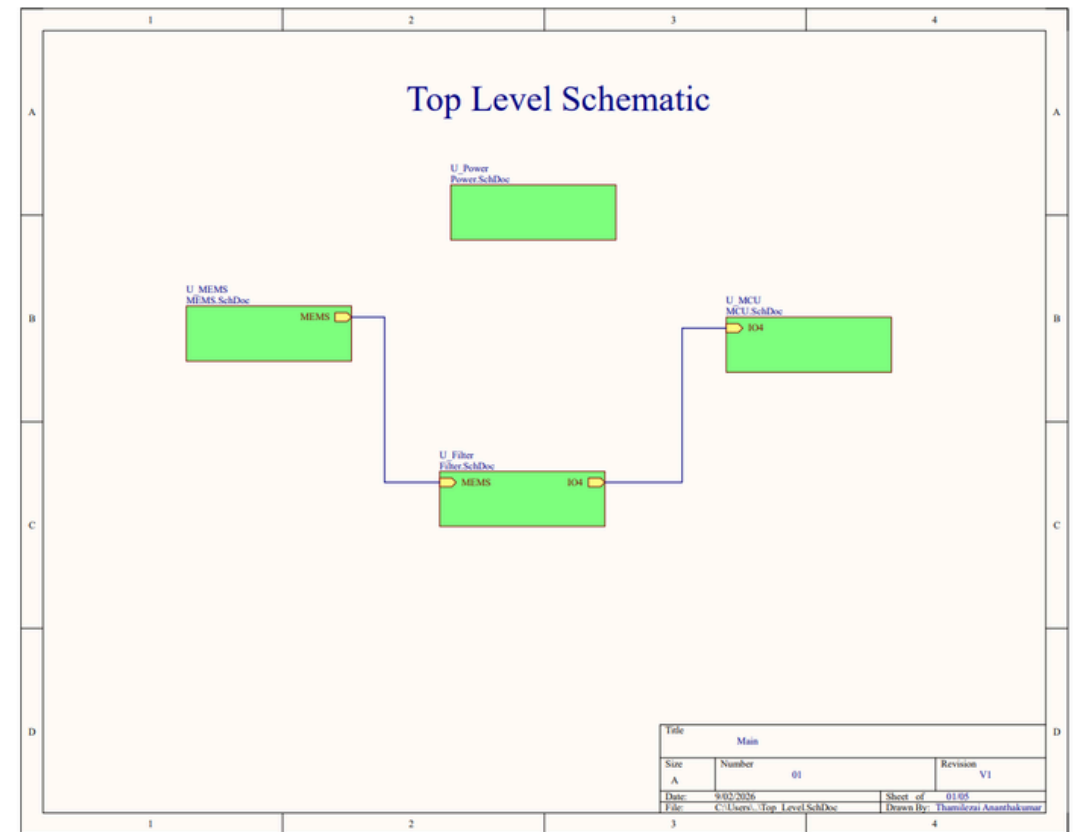
AC sweep of the final output: approx. 26 dB flat gain from 1 Hz to approx. 300 Hz, rolling off at higher frequencies.

## Scientific Explanation

- **Amplification:** U4 provides a gain of  $20\times$  ( $\approx 26$  dB), increasing the small PCG input to a usable amplitude.
- **Filtering:** The R7/R8–C4/C5 RC network attenuates frequencies above approximately 600 Hz, reducing high-frequency noise.
- **Biasing:** The R1/R2 divider with U6 generates a stable 1.65 V bias, centring the AC signal within the amplifier’s linear range.
- **Signal fidelity:** The final output n004 preserves the input’s waveform envelope and timing, indicating minimal distortion of the PCG characteristics.

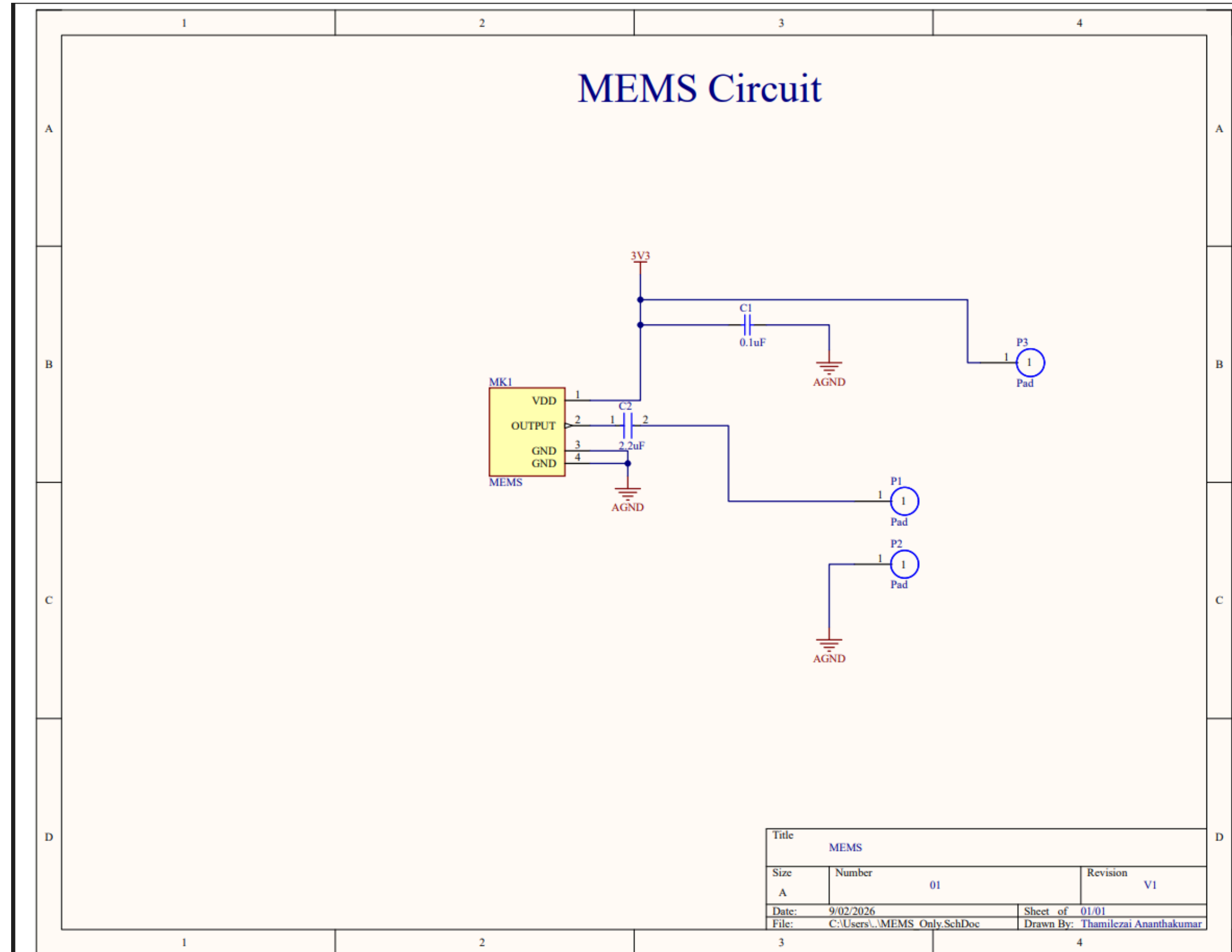


# Schematic Design : Main Board



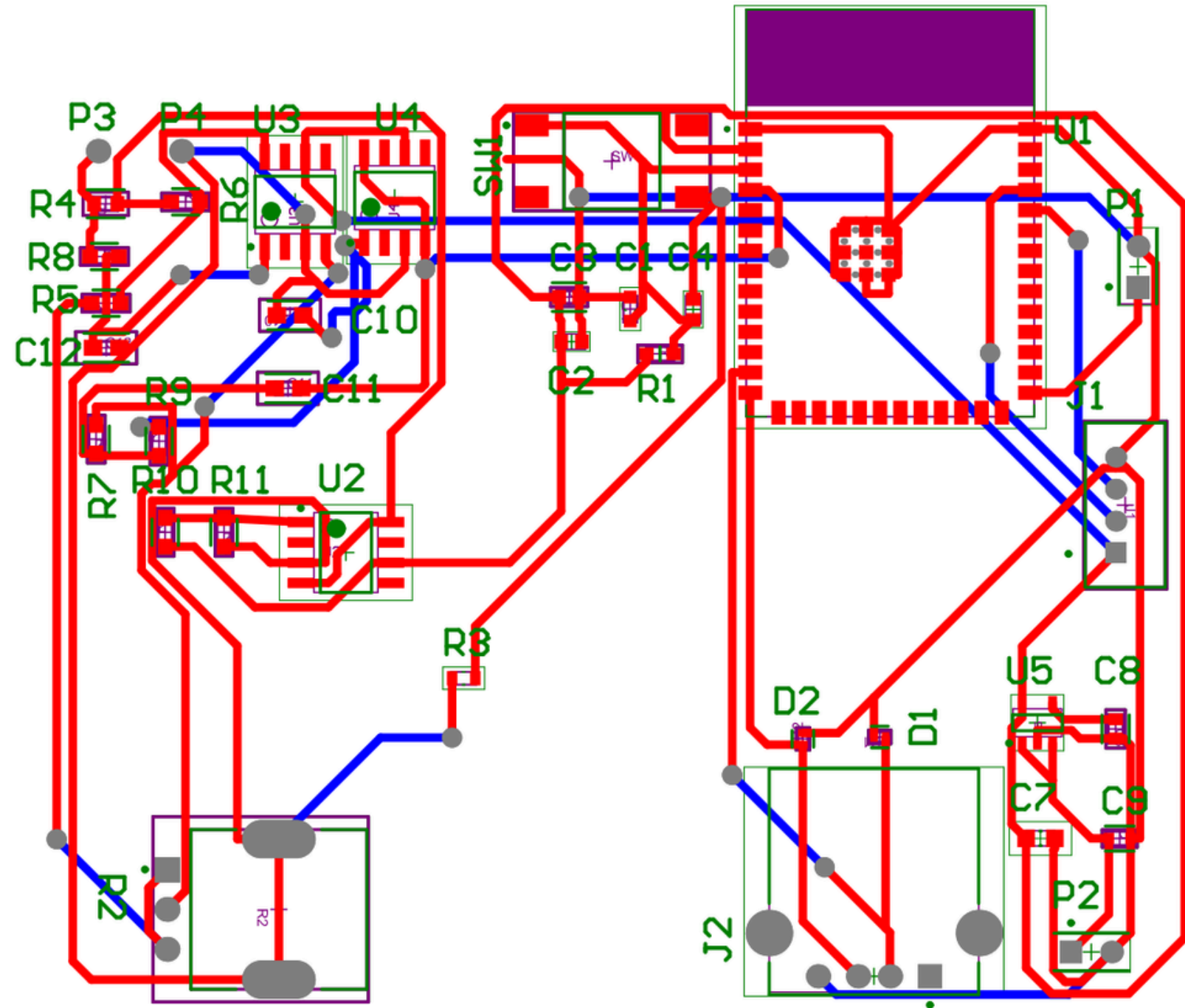


# Schematic Design : MEMS Board

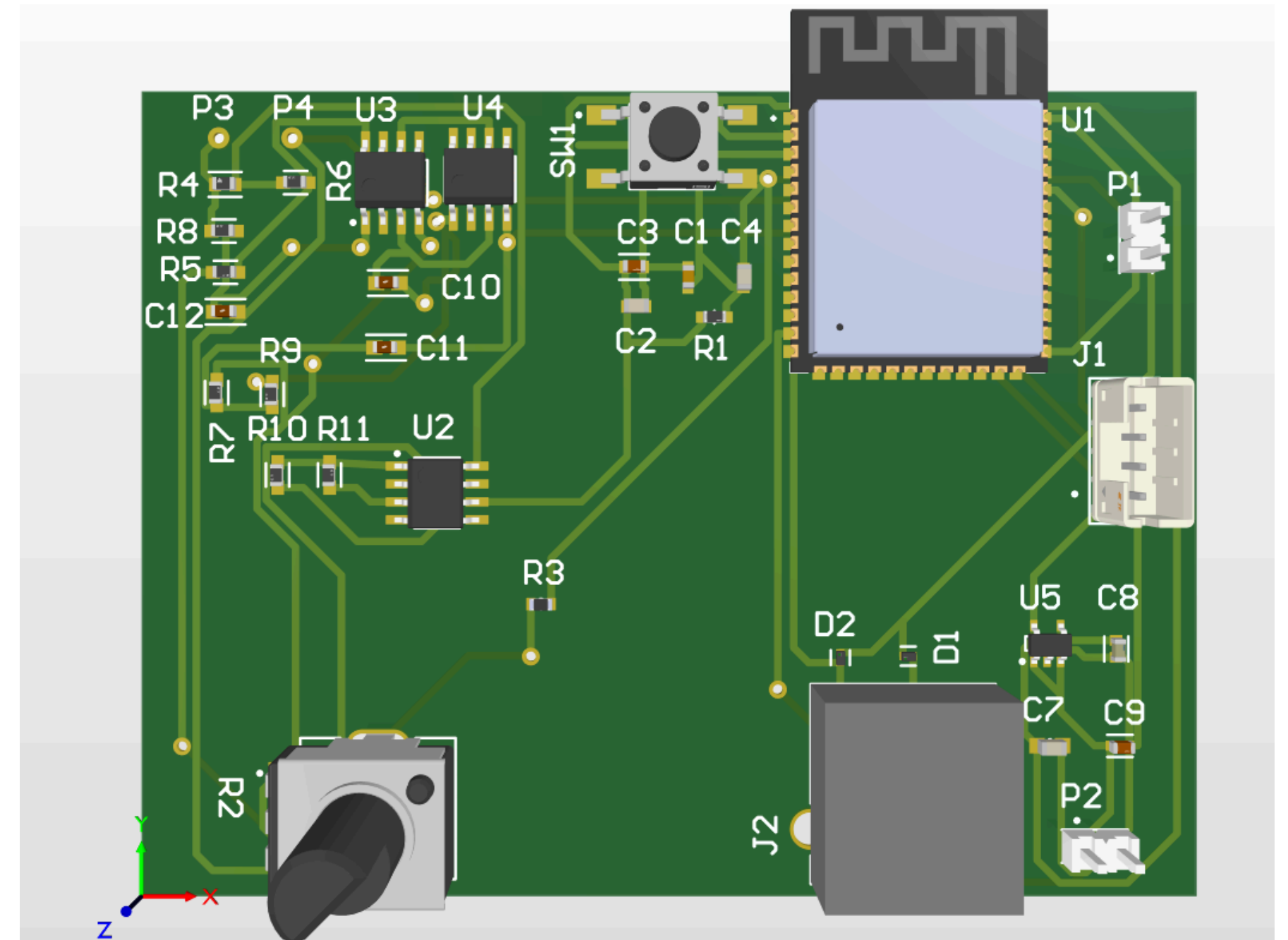




# PCB Design : Main Board



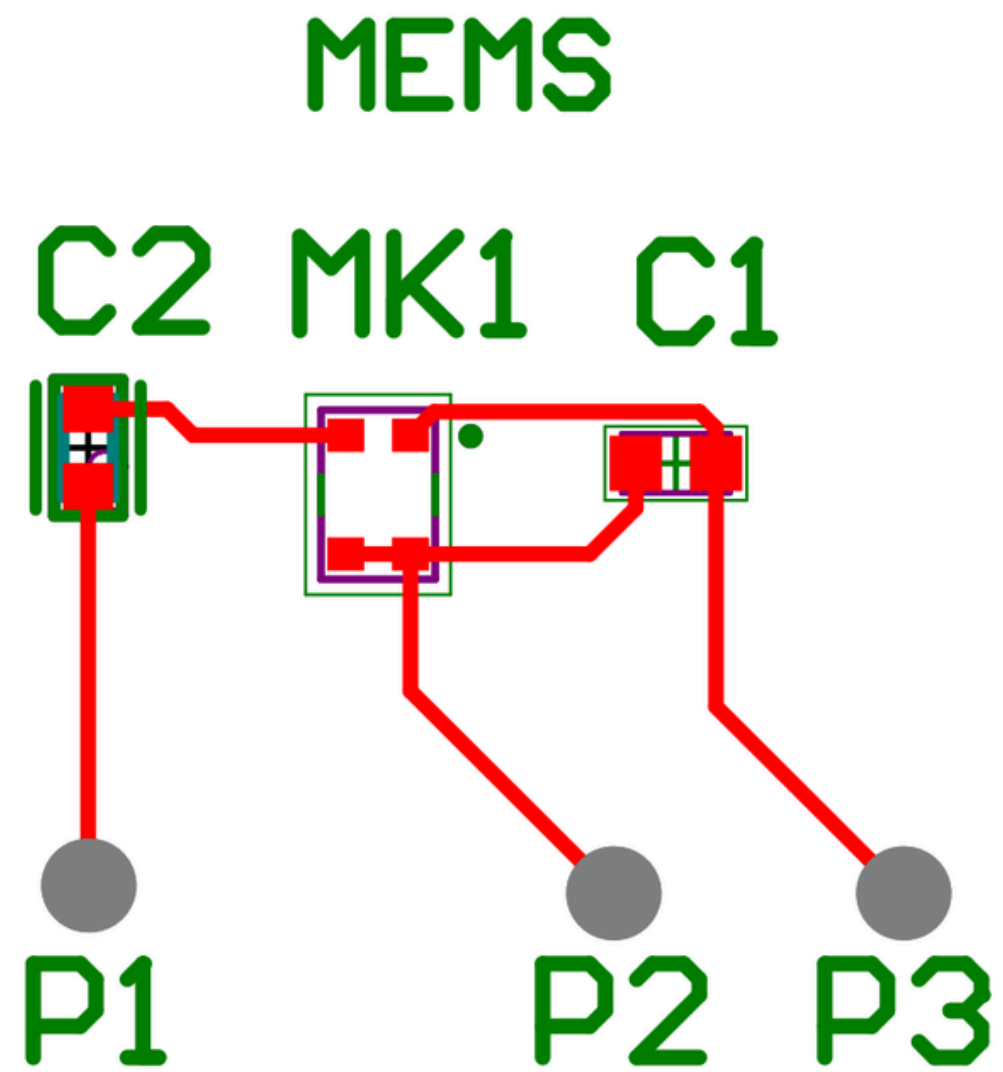
Routing



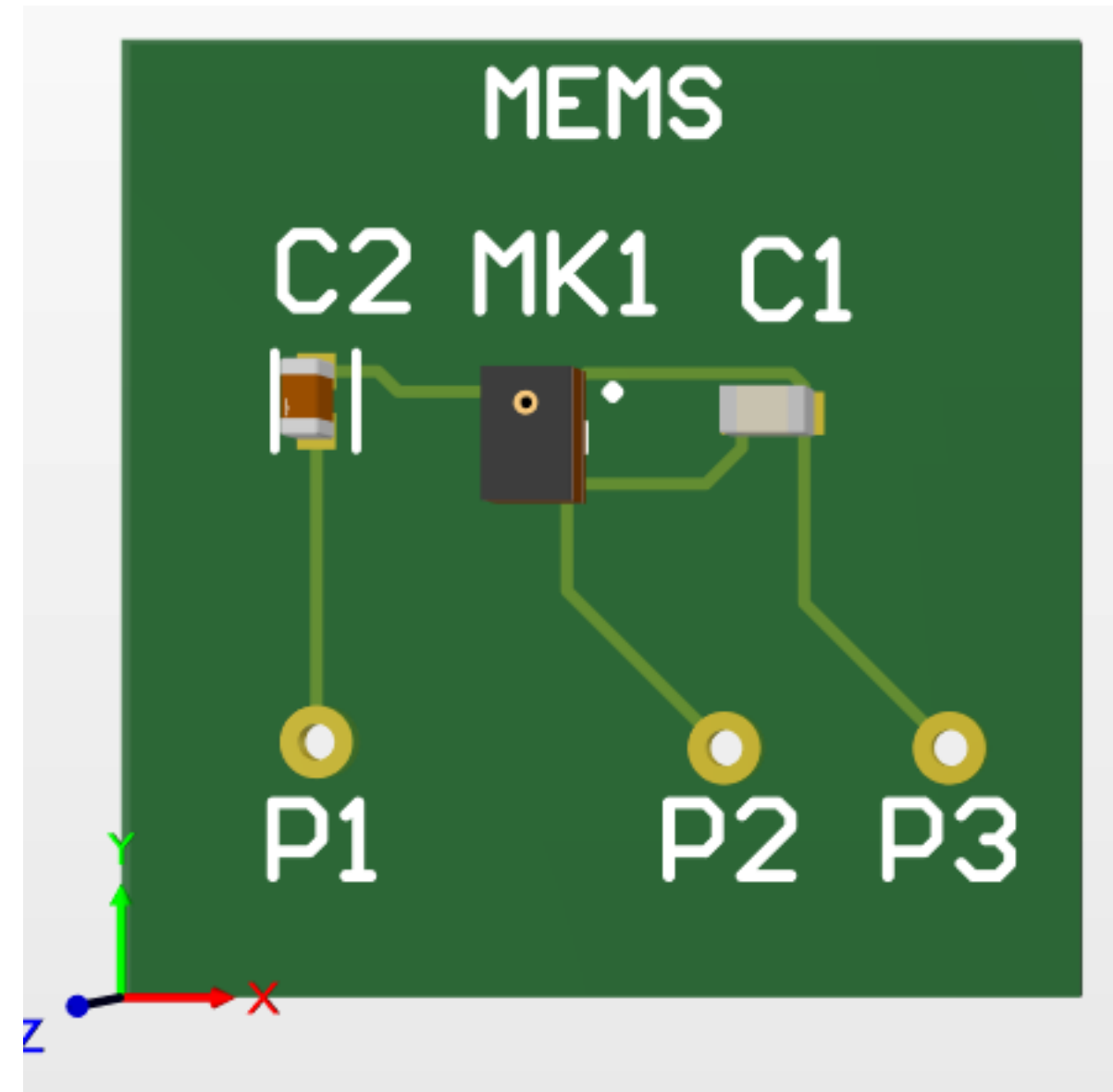
3D View



# PCB Design : MEM Board



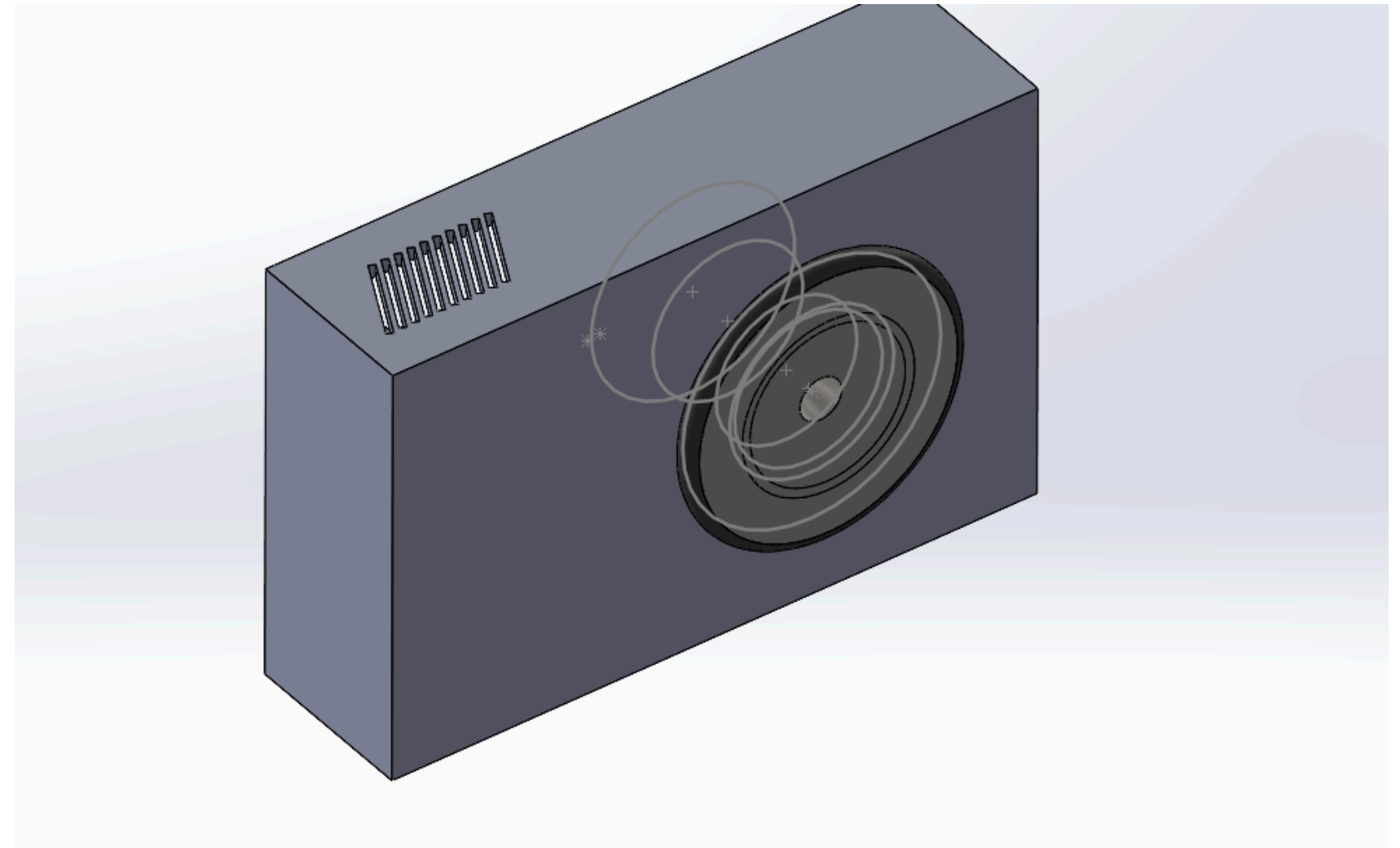
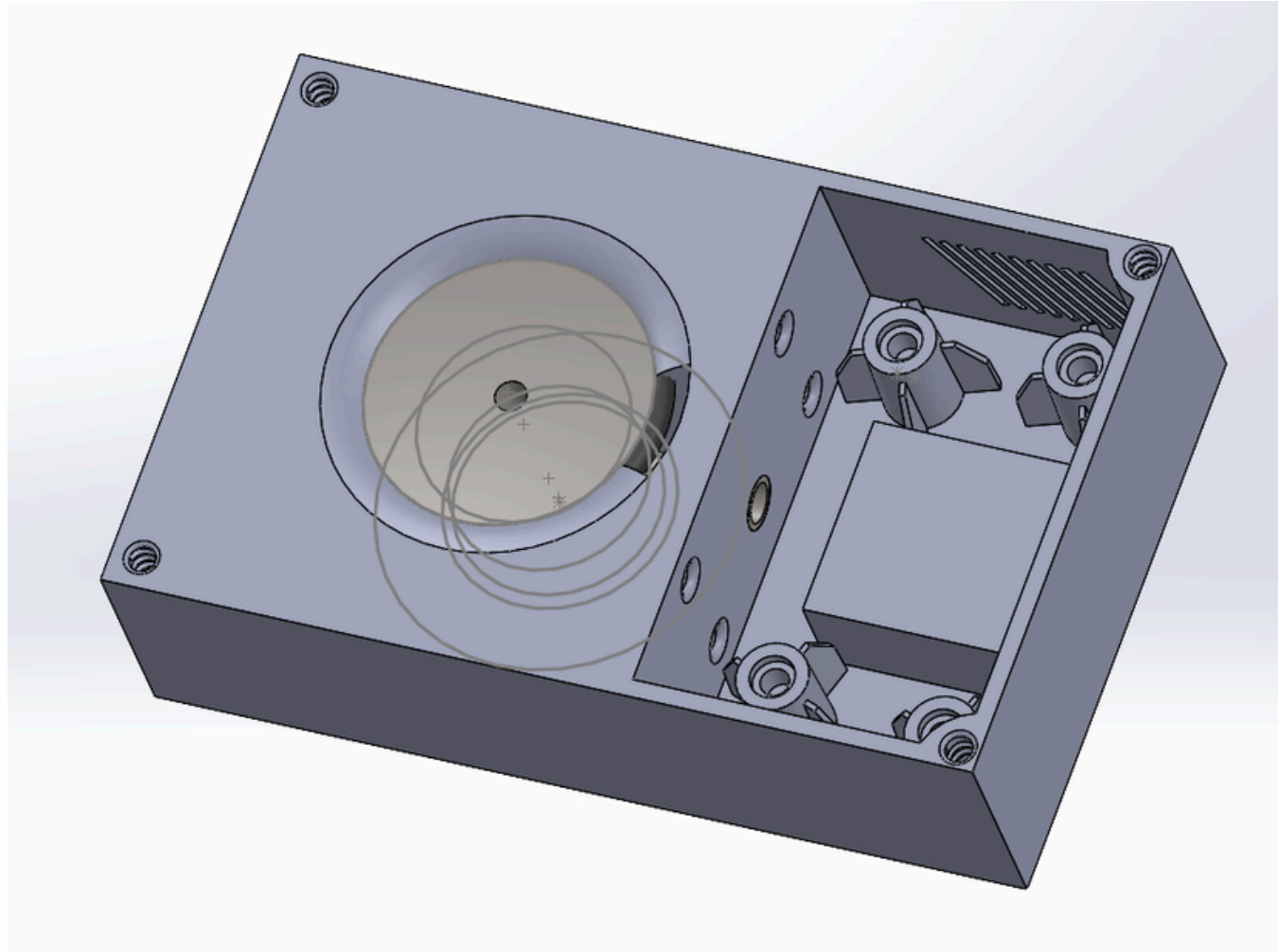
Routing



3D View



# Enclosure Design : Initial

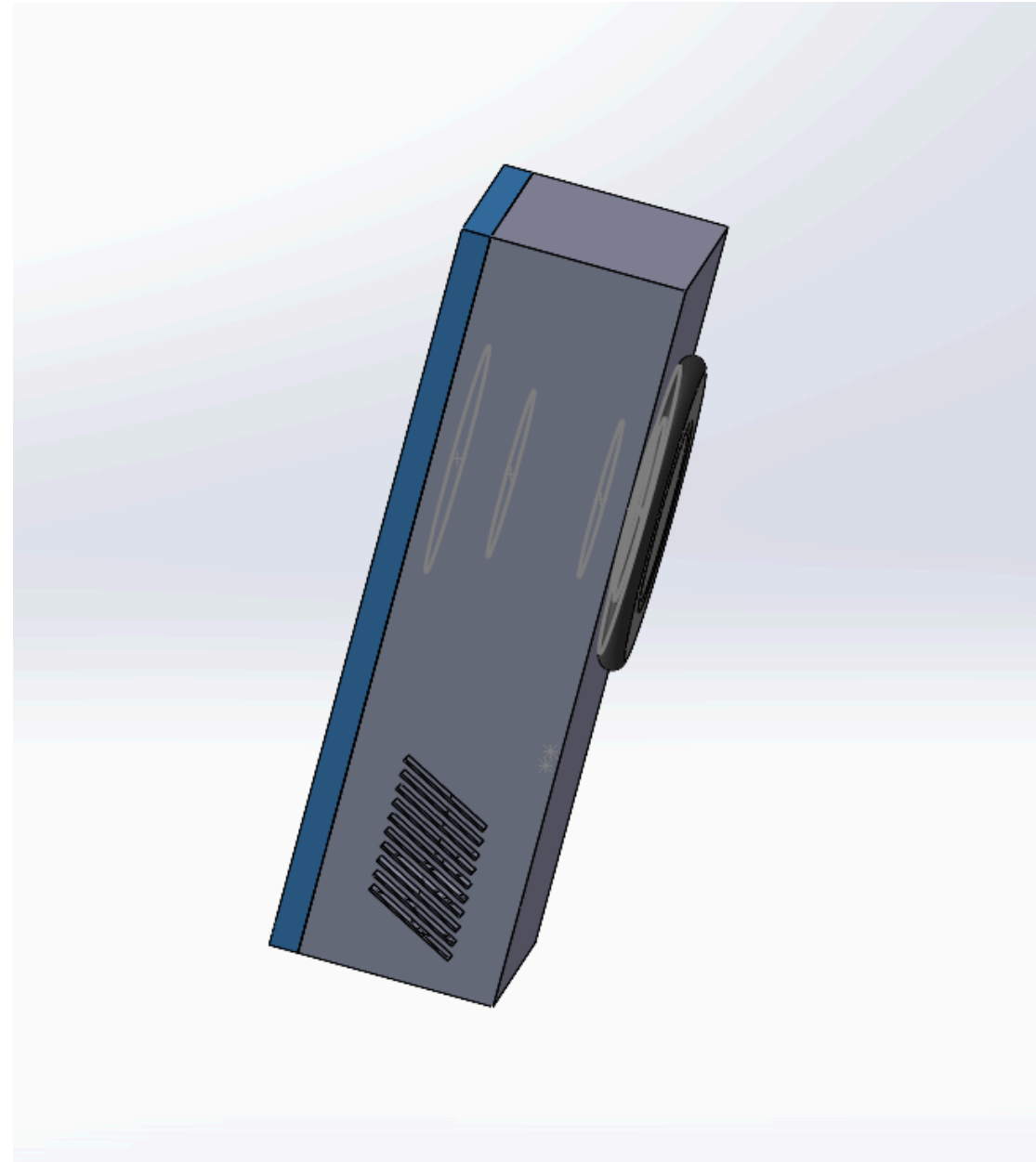


Bottom part



# Enclosure Design : Initial

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full design

